

Integral Relations among Totient Sections

Finite-level bases and bounded residue series

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ABSTRACT

We determine a basis and all integral relations for the sequences $n \mapsto \varphi(k^j n + r)$ with $0 \leq j \leq e$ and $0 \leq r < k^j$, where $k \geq 2$ and $e \geq 1$. The basis has $k^e + 1$ elements, and every other sequence is an explicit integer multiple of one of them. We prove independence by constructing nonsingular evaluation matrices and compare this argument with Martin's stronger affine theorem. A related congruence construction classifies rational binary series formed from dyadic totient residues. Irrationality of the unreduced binary totient series remains open.

1 Introduction

Splitting the totient sequence into its even and odd subsequences, then splitting once more, gives seven sequences:

$$\varphi(n); \quad \varphi(2n), \varphi(2n+1); \quad \varphi(4n), \varphi(4n+1), \varphi(4n+2), \varphi(4n+3).$$

The identities

$$\varphi(4n) = 2\varphi(2n), \quad \varphi(4n+2) = \varphi(2n+1)$$

reduce this list to five sequences, which are linearly independent over $\mathbb{Q}$. Every rational linear identity among the seven therefore follows from these two substitutions. The substitutions also preserve integral coefficients, so they describe the integral relations as well.

The same description holds at every finite depth and in every base. For an integer $k \geq 2$, the *totient sections* are the sequences $n \mapsto \varphi(k^j n + r)$, where $j \geq 0$ and $0 \leq r < k^j$. Throughout, n ranges over $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\varphi(0) = 0$. To describe relations we retain a separate index for each pair (j, r) , even when two pairs give the same sequence. For example, $\varphi(4n+2)$ and $\varphi(2n+1)$ are equal sequences but occupy different positions in the list above.

The sections form the k -kernel of the totient sequence. A sequence is k -regular when the $\mathbb{Z}$ -module generated by its k -kernel is finitely generated [1]. Coons proved that the totient sequence is not k -regular for any $k \geq 2$ [2, Thm. 3.2]. We describe the finitely

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generated modules obtained by stopping at a fixed level. The theorem below specifies a basis and expresses each remaining section as a scalar multiple of a basis element; these expressions give a test for every linear identity at that level. The level-zero family contains only $\varphi(n)$ and has dimension one.

Theorem 1.1 (A basis through each finite level). *Let $k \geq 2$ and $e \geq 1$ be integers, write $F_{j,r}^{(k)}(n) = \varphi(k^j n + r)$, and put*

$$V_{k,e} = \text{span}_{\mathbb{Q}} \{ F_{j,r}^{(k)} : 0 \leq j \leq e, 0 \leq r < k^j \}.$$

Then $\dim_{\mathbb{Q}} V_{k,e} = k^e + 1$, and

$$\mathcal{B}_{k,e} = \{ F_{0,0}^{(k)}, F_{1,0}^{(k)} \} \cup \{ F_{j,r}^{(k)} : 1 \leq j \leq e, 1 \leq r < k^j, k \nmid r \}$$

is a basis. Every omitted section reduces to a basis element by an explicit scalar: $F_{j,0}^{(k)} = k^{j-1} F_{1,0}^{(k)}$ for $j \geq 1$, and if $r = k^t u$ with $t = \max\{s : k^s \mid r\} \geq 1$ and $k \nmid u$, then

$$F_{j,r}^{(k)} = C_k(t, u) F_{j-t, u}^{(k)}, \quad C_k(t, u) = k^t \prod_{\substack{p \mid k \\ p \nmid u}} \left(1 - \frac{1}{p}\right).$$

The condition $k \nmid r$ allows residues with $\gcd(k, r) > 1$. For instance, $r = 1$ and $r = 2$ are both retained in base six, and the reductions involving them are

$$\varphi(36n + 6) = 2\varphi(6n + 1), \quad \varphi(36n + 12) = 4\varphi(6n + 2).$$

The multipliers differ because 2 divides $6n + 2$ but not $6n + 1$. The product formula also allows repeated prime factors of k , so there is no squarefreeness assumption on the base.

Martin's 2006 theorem on simultaneous totient inequalities already implies independence for every finite family of pairwise nonproportional affine forms with positive integral slopes [4, Thm. 1]. Its conclusion is stronger than the independence used here. The point of the present description is the explicit reduction of the sections and the integral relation basis in Corollary 3.1. We also give a direct proof that constructs a nonsingular evaluation matrix. It uses the CRT–Dirichlet separation argument in Yazdani's proof of Theorem 2, credited there to Shallit [12, pp. 652–653]. The step that needs care is to keep the progression for the selected affine form reduced while imposing prime divisors on all the others.

Section 2 proves the basis theorem. Section 3 starts with the six reductions in base six and derives the integral relation basis. In Section 4 we use prime isolation again, for bounded functions of totient residues. The series

$$S = \sum_{n \geq 1} \frac{\varphi(n)}{2^n}$$

has unbounded coefficients. Its irrationality, asked in Erdős Problem #249 [10, p. 61], remains open. The coefficient basis supplies no irrationality conclusion for this value. Appendix A gives further coefficient consequences; Appendices B and C collect the tail identities, comparison sequences and still-unproved cancellation estimates relevant to S . Their longer proofs are in *The Binary Totient Series*.

2 Proof of the basis theorem

Scalar reductions

There are two parts to the proof. The product formula reduces every omitted section to a retained one; independence then shows that no further reductions are possible. In the first part, the multiplier depends only on which prime divisors of k divide the argument. For a zero residue, kn and $k^j n$ have the same prime divisors when $n > 0$. Hence $\varphi(k^j n) = k^{j-1} \varphi(kn)$; both sides vanish at $n = 0$. For an omitted positive residue $r = k^t u$ with $t \geq 1$ and $k \nmid u$, we have $t < j$ because $0 < r < k^j$. Put $m = k^{j-t} n + u$. A prime $p \mid k$ divides m exactly when it divides u . The product formula for $\varphi(k^t m)/\varphi(m)$ therefore contributes k^t and one factor $1 - 1/p$ for each prime $p \mid k$ absent from u . The multiplier is an integer: its denominator is a product of distinct primes dividing k , and therefore divides k^t . This gives $C_k(t, u)$ and proves that the stated family spans $V_{k,e}$.

Affine independence

Independence follows from Martin's theorem. For any ordering of pairwise nonproportional affine forms and any $C > 0$, it gives a set of positive lower density on which each successive totient ratio exceeds C . In a proposed nonzero relation, place a nonzero coefficient c_1 first and take $C > \max(1, \sum_{i>1} |c_i|/|c_1|)$. The first term then dominates the sum of all remaining terms in absolute value, a contradiction [4, Thm. 1]. This also proves independence after any finite prefix is deleted.

We give a direct proof by constructing a square matrix of evaluations with nonzero determinant. Let $L_i(n) = a_i n + b_i$ for $1 \leq i \leq s$, where $a_i > 0$ and $a_i b_j - a_j b_i \neq 0$ for $i \neq j$. A shift of n makes all the b_i positive and leaves these cross determinants unchanged. Write $g_i = \gcd(a_i, b_i)$ and $A_i = L_i/g_i$. If $g_i > 1$, the values of L_i have a fixed divisor, so it is the primitive form A_i , not L_i , that we will make prime. We will choose a prime ℓ and points $n_1, \dots, n_s$ for which

$$E = (\varphi(L_j(n_i)))_{1 \leq i, j \leq s} \equiv \begin{pmatrix} \varphi(g_1) & 0 & \cdots & 0 \\ 0 & \varphi(g_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \varphi(g_s) \end{pmatrix} \pmod{\ell}, \quad \ell \nmid \prod_i \varphi(g_i). \quad (1)$$

A rational relation among the sequences would give a vector in the kernel of E . The displayed congruence makes its determinant nonzero, so no such relation can exist.

For row i , we require $L_j(n_i)$ to have a prime divisor $q_{ij} \equiv 1 \pmod{\ell}$ whenever $j \neq i$. Then $q_{ij} - 1$ divides $\varphi(L_j(n_i))$, giving the off-diagonal zeros in (1). For the diagonal entry, we require $A_i(n_i)$ to be a prime congruent to 2 modulo ℓ . Choose an odd ℓ dividing none of the slopes a_i , the products $g_i \varphi(g_i)$ or the nonzero cross determinants. For each fixed i , Dirichlet's theorem supplies distinct primes $q_{ij} \equiv 1 \pmod{\ell}$ ($j \neq i$) outside the same finite exceptional set. We impose

$$A_i(n) \equiv 2 \pmod{\ell}, \quad L_j(n) \equiv 0 \pmod{q_{ij}} \quad (j \neq i).$$

Each congruence is soluble because its slope is coprime to its modulus. CRT combines them into $n = n_0 + Ru$, where $R = \ell \prod_{j \neq i} q_{ij}$.

Dirichlet's theorem must now be applied to the values of A_i , rather than to n . These values form the progression

$$A_i(n_0 + Ru) = A_i(n_0) + (a_i/g_i)Ru.$$

It remains to check that its initial value and step are coprime. Primitivity excludes the prime factors of a_i/g_i , and the residue 2 excludes the odd prime ℓ . A prime q_{ij} cannot divide $A_i(n_0)$: it would then divide both $L_i(n_0)$ and $L_j(n_0)$, hence

$$a_i L_j(n_0) - a_j L_i(n_0) = a_i b_j - a_j b_i,$$

contrary to its choice. This accounts for every prime factor of the step. We may therefore choose $p = A_i(n_i)$ prime, as large as needed, and in particular with $p \nmid g_i$. Then

$$\varphi(L_i(n_i)) = \varphi(g_i)(p-1) \equiv \varphi(g_i) \not\equiv 0 \pmod{\ell},$$

while all the other entries of row i vanish modulo ℓ . We choose the rows separately, applying Dirichlet's theorem once more for each selected form A_i . Thus no simultaneous primality assertion is required. The construction proves (1) with every n_i beyond any prescribed threshold.

The two zero-residue sections

The affine-independence argument does not separate $\varphi(n)$ and $\varphi(kn)$, because their arguments are proportional. We use two different values of their ratio. On $n = km + 1$ the ratio is $\varphi(k)$, so a proposed relation contains only one zero-residue term after this restriction. The positive-residue sections become the forms

$$k^{j+1}m + k^j + r \quad (k \nmid r),$$

together with $km + 1$. These are pairwise nonproportional: equality of two intercept-to-slope ratios would give $r/k^j = r'/k^{j'}$, which for $j' > j$ forces $k \mid r'$. No positive-residue form is proportional to $km + 1$. Affine independence eliminates every positive-residue coefficient and leaves $A + B\varphi(k) = 0$, where A, B multiply $\varphi(n), \varphi(kn)$. We now evaluate the remaining relation at $n = k$ to obtain $A + Bk = 0$. Since $\varphi(k) < k$, the two equations force $A = B = 0$. Finally,

$$|\mathcal{B}_{k,e}| = 2 + \sum_{j=1}^e (k^j - k^{j-1}) = k^e + 1,$$

which completes the proof of Theorem 1.1. For independence on a tail, use a sufficiently large power of k in the last evaluation.

3 Integral relations

The reductions give a second way to test a linear identity, by substitution. Consider base six at level two, where the family has 43 indices. The 37 retained sections are $F_{0,0}, F_{1,0},$

the five $F_{1,r}$ ($1 \leq r \leq 5$), and the 30 sections $F_{2,r}$ with $1 \leq r < 36$ and $6 \nmid r$. The omitted sections satisfy

$$\begin{aligned} F_{2,0} &= 6F_{1,0}, & F_{2,6} &= 2F_{1,1}, & F_{2,12} &= 4F_{1,2}, \\ F_{2,18} &= 3F_{1,3}, & F_{2,24} &= 4F_{1,4}, & F_{2,30} &= 2F_{1,5}. \end{aligned}$$

Thus an integral combination of the 43 sections vanishes precisely when these six substitutions leave all 37 retained coefficients zero. To recover the relation itself, keep track of what each substitution removes. Let $E_{j,r}$ denote the generator at index (j,r) in the free abelian group on all section indices. The substitution $F_{2,12} = 4F_{1,2}$ corresponds to the decomposition

$$x_{1,2}E_{1,2} + x_{2,12}E_{2,12} = (x_{1,2} + 4x_{2,12})E_{1,2} + x_{2,12}(E_{2,12} - 4E_{1,2}).$$

The first coefficient tests whether the retained component vanishes; the second records the multiple of the relation removed. Doing this at all six omitted indices gives every integral relation, since the retained sections are independent. At base two and level two the relations are $E_{2,0} - 2E_{1,0}$ and $E_{2,2} - E_{1,1}$. The following corollary expresses the same calculation at arbitrary base and level.

Corollary 3.1 (Integral coordinates and all integral relations). *Let $k \geq 2$ and $e \geq 1$. The retained family is a $\mathbb{Z}$ -basis of the module generated by the sections through level e . Index the sections by their level and residue, retaining distinct indices even when they define equal sequences. For each omitted index i , write the scalar reduction as $F_i = a_i F_{j(i)}$, where $j(i)$ is retained and a_i is a nonnegative integer. In the free abelian group with one generator E_i for each of these indices, the vectors*

$$R_i = E_i - a_i E_{j(i)} \quad (i \text{ omitted})$$

form a $\mathbb{Z}$ -basis of the kernel of evaluation $E_i \mapsto F_i$. In particular, every integral relation has a unique integral expression in these elementary relations, and their rank is $\sum_{j=1}^{e-1} k^j$.

Proof. Since $\prod_{p|k, p \nmid u} p$ divides k , every scalar $C_k(t, u)$ is an integer. The retained family therefore spans the section module over $\mathbb{Z}$, and its rational independence makes it a $\mathbb{Z}$ -basis.

For any vector $x = \sum_i x_i E_i$, substitute $E_i = R_i + a_i E_{j(i)}$ at each omitted index and collect terms:

$$x = \sum_{j \text{ retained}} \left(x_j + \sum_{\substack{i \text{ omitted} \\ j(i)=j}} a_i x_i \right) E_j + \sum_{i \text{ omitted}} x_i R_i.$$

Evaluation kills the second sum. Independence of the retained sections therefore says that x is a relation exactly when every coefficient in the first sum is zero. In that case the second sum is its expression in the R_i . This expression is unique: the coefficient at omitted index i in $\sum_h c_h R_h$ is c_i itself. The unit coefficient at that index is also why no division is needed. Counting omitted indices gives $(1 + k + \dots + k^e) - (k^e + 1) = \sum_{j=1}^{e-1} k^j$. $\square$

The assertion concerns the module generated by the sections. It need not contain every integer-valued sequence in its rational span: for $k = 2$ and $e \geq 2$, the sequence $\varphi(4n + 3)/2$ is integer-valued and has coefficient $1/2$ in the retained basis. Thus this module is not saturated in the group of integer-valued sequences.

An alternative is to test equality at finitely many inputs. Return to base six and level two. Order the retained columns as above, with residues increasing within each level, and evaluate at $n = 0, \dots, 36$ to obtain a 37×37 matrix. Its determinant is 17 modulo 101, so it is nonsingular over $\mathbb{Q}$. Consequently a retained combination is zero if and only if all 37 of its integer evaluations are zero. Here reduction modulo 101 certifies the matrix, not the equality of a proposed combination: $101F_{2,2}$ would vanish modulo 101 at every input, although its integer value at $n = 0$ is 101. The script `scripts/totient_kernel_normal_form.py` calculates the coordinates and the first nonzero evaluation using integer arithmetic.

4 Bounded functions of totient residues

A similar use of congruences produces a nonzero totient residue between arbitrarily long blocks of zero residues. We use this to classify the rational series obtained from bounded functions of dyadic totient residues. The proof rests on a tail estimate: one zero block makes the tail vanish under the assumption of rationality, and the other leaves a nonzero term too small for that assumption. In the first sum below, residues are represented by their least nonnegative values. These coefficients need not be binary digits.

Theorem 4.1 (Series of totient residues). *For every $m \geq 3$,*

$$\sum_{n \geq 1} \frac{\varphi(n) \bmod m}{2^n} \notin \mathbb{Q}.$$

For $k \geq 1$ and $f : \mathbb{Z}/2^k\mathbb{Z} \rightarrow \mathbb{Q}$, the series $\sum_{n \geq 1} f(\varphi(n) \bmod 2^k)2^{-n}$ is rational exactly when f is constant on the even residue classes. If that constant is c , its value is $3f(1)/4 + c/4$.

Modulo two the sum is $3/4$, since $\varphi(n)$ is even for $n \geq 3$. More generally only $f(1)$ and the even residue classes occur in the second assertion. The following coefficient pattern is the basis of the proof:

$$\underbrace{0, \dots, 0}_{L \text{ terms}}, \quad a_N \neq 0, \quad \underbrace{0, \dots, 0}_{L \text{ terms}}.$$

The coefficients outside these blocks may have either sign. Their uniform bound is what makes the following argument possible.

Lemma 4.2 (An isolated nonzero coefficient between long zero blocks). *Let $a_n \in \mathbb{Z}$ satisfy $|a_n| \leq C$. Suppose that for arbitrarily large L there is $N > L$ such that $a_N \neq 0$ and $a_{N+t} = 0$ for $0 < |t| \leq L$. Then $\sum_{n \geq 1} a_n 2^{-n}$ is irrational.*

Proof. Suppose that the sum has denominator $q \geq 1$. Multiplying it by 2^j and removing its finite integer prefix shows that every scaled tail $T_j = \sum_{t \geq 1} a_{j+t} 2^{-t}$ belongs to $q^{-1}\mathbb{Z}$. Choose a pattern with $C2^{-L} < 1/q$. The block on the right gives

$$|T_N| \leq C \sum_{t > L} 2^{-t} = C2^{-L} < 1/q.$$

The only element of $q^{-1}\mathbb{Z}$ in this interval is zero, so $T_N = 0$. Now start before the left block. The next L coefficients vanish, and the entire contribution after a_N has just been shown to be zero. Consequently

$$T_{N-L-1} = a_N 2^{-L-1}, \quad 0 < |T_{N-L-1}| \leq C 2^{-L-1} < 1/q,$$

contradicting $T_{N-L-1} \in q^{-1}\mathbb{Z}$. $\square$

For example, the indicator of the squares satisfies the lemma. An eventually periodic sequence with infinitely many nonzero terms does not have isolated coefficients at arbitrarily large distances. The uniform bound C is essential to this argument.

Proof of Theorem 4.1. Fix a unit $s \pmod{m}$ and a positive integer L . Choose distinct primes $q_t \equiv 1 \pmod{m}$, all exceeding L , for $0 < |t| \leq L$. The congruences

$$p \equiv s \pmod{m}, \quad p \equiv -t \pmod{q_t} \quad (0 < |t| \leq L)$$

define a reduced progression, since s is a unit modulo m and $0 < |t| < q_t$. CRT and Dirichlet's theorem therefore allow us to choose an arbitrarily large prime $p > L + 1$ on this progression, with $m \mid \varphi(p + t)$ at every neighbour and $\varphi(p) \equiv s - 1 \pmod{m}$ at the centre.

For a rational-valued function f on the residue classes, subtract $f(0)$ and multiply by a common denominator of its values. Since $\sum_{n \geq 1} 2^{-n} = 1$, these operations preserve whether the sum is rational. If $f(s - 1) \neq f(0)$, the resulting bounded integer sequence has an isolated nonzero coefficient for every L , so the lemma applies. For the least-residue map we take $s = -1$ and obtain the nonzero residue $m - 2$ when $m \geq 3$. When $m = 2^k$, each even residue r is of the form $s - 1$ with s a unit, so nonconstancy on the even classes gives an irrational sum by the same argument. Conversely, if f is constant there, all terms after $n = 2$ equal c and we obtain $3f(1)/4 + c/4$. $\square$

The first assertion is also an application of Erdős's sparse-series criterion [6, Lem. 1, p. 213], and of Kaneko, Suzuki and Tachiya [7, Thm. 3]. The residues are bounded and nonnegative, and the prime construction above makes their support infinite. The nonzero residues occur on a set of density zero: an integer in that set has no prime divisor $p \equiv 1 \pmod{m}$. Wong gave the case in which the base equals the modulus [5]. The isolated-coefficient argument above also treats signed values of $f - f(0)$, for which those nonnegative sparse-series criteria need an additional spacing hypothesis. The [extension to other bases](#) treats every integer $t \geq 2$ and proves rational linear independence of 1 and $\sum_n (\varphi(n) \bmod m) t^{-dn}$, $d \geq 1$.

A Further consequences for the sections

The finite-level theorem also describes the full dyadic kernel. The notation below specialises the base to two.

Corollary A.1 (Dyadic basis). *The family*

$$\{\varphi_{0,0}, \varphi_{1,0}\} \cup \{\varphi_{j,r} : j \geq 1, 0 < r < 2^j, r \text{ odd}\}, \quad \varphi_{j,r}(n) = \varphi(2^j n + r),$$

is a basis for the rational span of the full dyadic kernel. Every rational relation is generated by the scalar reductions. The complete level-zero truncation has dimension one.

Proof. Each finite subset of this family is contained in some $\mathcal{B}_{2,e}$. Theorem 1.1 gives independence, and its reductions give spanning. At level zero the sole section $\varphi(n)$ is nonzero, since $\varphi(1) = 1$. $\square$

The affine argument also permits periodic coefficients by restriction to their common residue classes.

Corollary A.2 (Periodic coefficients). *Let $L_1, \dots, L_s$ be pairwise nonproportional affine forms with integer coefficients and positive slopes. If $w_1, \dots, w_s$ are rational-valued periodic sequences, then*

$$\sum_i w_i(n) \varphi(L_i(n)) = 0 \quad \text{for all sufficiently large } n \quad \implies \quad w_i(n) = 0 \quad \text{for every } i, n.$$

Proof. We choose a common period Q and restrict to $n = r + Qt$. The coefficients are now the constants $w_i(r)$, and the cross determinants are $Q(a_i b_j - a_j b_i) \neq 0$. By shifting t beyond the given threshold we can apply affine independence, obtaining $w_i(r) = 0$ for every i . We repeat this for each residue r ; periodicity then gives the conclusion even at the indices preceding the threshold. $\square$

Proportional forms are excluded for a reason: the identity $\varphi(4n + 2) = \varphi(2n + 1)$ would otherwise contradict the conclusion. Martin's theorem gives the same corollary on each progression [4, Thm. 1].

Coons's proof of nonregularity counts the poles of $\zeta(s - 1)/\zeta(s)$ [2, Cor. 3.1, Thm. 3.2]. For an integer-valued sequence, finite dimension of the rational kernel span is equivalent to finite generation of its integral kernel module: choose finitely many evaluations that embed the span in $\mathbb{Q}^d$; the module embeds in $\mathbb{Z}^d$, whose subgroups are finitely generated (compare [3, App. A]). Theorem 1.1 recovers this nonregularity and gives the ranks of its finite truncations. For an account of the earlier separation method, see Allouche, Shallit and Yassawi [13, Thm. 3, Ex. 4, Rem. 7].

These facts have a functional consequence. A k -Mahler equation has the form $\sum_{i=0}^d P_i(z) F(z^{k^i}) = 0$, with $P_i \in \mathbb{Q}[z]$ and $P_0 \neq 0$. The height theorem of Adamczewski, Bell and Smertnig implies that a k -Mahler series with integer coefficients of polynomial growth is k -regular [16, Thm. 1.2(b)]. Since $\varphi(n) \leq n$, its logarithmic height is at most $\log n$, so $F_\varphi(z) = \sum_{n \geq 1} \varphi(n) z^n$ has no such equation. Bell and Smertnig's multiplicative classification gives the same conclusion [14, Thm. 1.3]. Bell, Bruin and Coons treat related algebraic and D -finite series, the latter satisfying a linear differential equation with polynomial coefficients [15, Thms. 1.5–1.6]. These statements concern the function, and do not determine $F_\varphi(1/2)$.

The section operators $T_i g(n) = g(kn + i)$ satisfy

$$T_i F_{j,r} = F_{j+1, k^j i + r}, \quad V_{k,e+1} = V_{k,e} + \sum_{i=0}^{k-1} T_i V_{k,e}.$$

To obtain the second identity, write every residue at level $j + 1$ as $k^j i + r$. For a positive retained residue, the resulting section is again retained. For the two zero-residue generators, use $T_i F_{0,0} = F_{1,i}$ and $T_i F_{1,0} = F_{2,ki}$ followed by the zero-residue or scalar reduction. We thus obtain a positive integer multiple of a basis element in every case, although the increasing dimensions prevent any finite truncation from being invariant. The growth formula using the joint spectral radius for finite-dimensional regular sequences [3, Thm. 1, Prop. 4, Cor. 7] therefore does not apply to φ .

B The binary totient series

The proof of Theorem 4.1 uses boundedness where a long zero block makes the tail small. For S , the estimate $0 \leq \varphi(n) \leq n$ proves convergence, but forcing $2^L \mid \varphi(N + L)$ with $N + L > 1$ already requires $2^L \leq N + L - 1$. The tail bound $(N + L + 2)2^{-L}$ then exceeds 1. Thus this divisibility construction does not supply the small nonzero tail needed to exclude a rational value. We consider instead a finite test for differences of scaled tails.

Put $R_N = \sum_{j \geq 1} \varphi(N + j)2^{-j}$ and $\Delta_h(N) = R_{N+h} - R_N$. Subtracting the finite-prefix identities for $2^{N+h}S$ and $2^N S$ gives

$$\Delta_h(N) \equiv 2^N(2^h - 1)S \pmod{\mathbb{Z}}. \quad (2)$$

If $S = a/(2^c v)$, with v odd, then $\Delta_h(N)$ is integral for $N \geq c$ and $h = \varphi(v)$. To disprove rationality it therefore suffices to find a nonintegral difference beyond every threshold for each positive h .

For $h, N, L \in \mathbb{N}$, define the integer

$$D_{h,N,L} = \sum_{j=0}^{L-1} (\varphi(N + h + 1 + j) - \varphi(N + 1 + j))2^{L-1-j}, \quad B = N + h + L + 2.$$

Since $0 \leq R_M \leq M + 2$, we have

$$2^L \Delta_h(N) - D_{h,N,L} = R_{N+h+L} - R_{N+L}, \quad |2^L \Delta_h(N) - D_{h,N,L}| \leq B. \quad (3)$$

The two tails are nonnegative, so their difference is bounded in absolute value by the larger tail bound, not the sum of the bounds. Thus $D_{h,N,L}/2^L$ approximates $\Delta_h(N)$ with error at most $B/2^L$. To exclude an integer value, the approximation must lie farther than this error from every integer. Equivalently, the residue of $D_{h,N,L}$ must avoid both ends of the interval $[0, 2^L)$. Thus the finite inequality

$$B < D_{h,N,L} \pmod{2^L} < 2^L - B \quad (4)$$

forces $\Delta_h(N) \notin \mathbb{Z}$. Conversely, hold h, N fixed and suppose that $\|\Delta_h(N)\|_{\mathbb{R}/\mathbb{Z}} > 0$. Choose L so that $2^L \|\Delta_h(N)\|_{\mathbb{R}/\mathbb{Z}} > 2B$: the left side grows exponentially, whereas $B = N + h + L + 2$ grows linearly. The same error estimate gives (4). This detects every nonintegral difference at some depth. A failed test at one depth proves no integrality: the error interval may simply be too wide to separate the value from $\mathbb{Z}$. Increasing the depth eventually detects each fixed nonintegral difference, but does not produce such a difference for each h . For example, $D_{1,12,16} = -143140$ has residue 53468 modulo 2^{16} , and $B = 31$, so it excludes denominators dividing 2^{12} .

B.1 Equivalent residue conditions

We write $K(h, N, L)$ for (4).

Theorem B.1 (Propagation of one nonintegral tail difference). *Fix $d \geq 1$ and $N \geq 0$. If $\Delta_d(N) \notin \mathbb{Z}$, then every sufficiently late pair $\{t, t+1\}$ contains an m such that $K(md, N, md)$ holds. Consequently*

$$\exists t \geq 1 : K(td, N, td) \iff \Delta_d(N) \notin \mathbb{Z}.$$

Requiring this for every $d \geq 1, N \geq 0$ is equivalent to $S \notin \mathbb{Q}$.

Here the depth is no longer free: it must equal the shift. The recurrence $\Delta_{(t+1)d}(N) \equiv 2^d \Delta_{td}(N) + \Delta_d(N) \pmod{\mathbb{Z}}$ prevents two consecutive large shifts from both failing the test. Two failures would force the fixed nonintegral $\Delta_d(N)$ arbitrarily close to an integer. The estimates are in [Appendix A of the companion paper](#). A single pair d, N need not imply irrationality; the final assertion requires every positive d and every $N \geq 0$.

For a hypothetical denominator $2^c v$, a second test uses the modulus $M = (2^H - 1)/v$, where H is a positive multiple of $\varphi(v)$. Write $B_{H,c} = \sum_{j=0}^{H-1} \varphi(c+1+j)2^{H-1-j}$.

Equivalent formulation B.2 (A residue test modulo a divisor of $2^H - 1$). The series S is irrational if and only if, for every $c \geq 0$ and odd $v \geq 1$, some positive multiple H of $\varphi(v)$ satisfies

$$c + H + 1 < (-B_{H,c}) \bmod M < M - (c + H + 1). \quad (5)$$

The choice of modulus makes $B_{H,c} = M(vR_c) - \Delta_H(c)$. Under the hypothetical rationality assumption, vR_c is an integer, so the first term is a multiple of M and only the small tail difference remains modulo M . The [proof and residue recurrence](#) also give the converse by keeping c, v fixed and letting H grow. The required residues have not been shown to exist for every choice of the parameters in either test. A separate [Farey-interval calculation](#) at depth 240 excludes $S = a/q$ for $0 < q \leq Q_0 \approx 7.96 \times 10^{34}$.

B.2 Comparison sequences

Large ranks alone cannot settle the series question. There is an integer sequence c with $0 \leq c(n) \leq n$, unchanged odd totients and $|c(n) - \varphi(n)| \leq 2$ at even indices, for which $\sum c(n)2^{-n} = 5/4$. Its carry is the integer sequence

$$u_N = 4 \sum_{j \geq 1} c(N+j)2^{-j} = 5 \cdot 2^N - 4 \sum_{n=1}^N c(n)2^{N-n}.$$

For $j \geq 1$ and odd $0 < r < 2^j$, the recurrence gives

$$4\varphi(2^j n + r) = 2u_{2^j n + r - 1} - u_{2^j n + r}.$$

Thus the $2^e - 1$ independent odd-residue totient sections lie in the span of the carry sections through level e . This explains how the rank lower bound survives even though the sum is rational. The [construction and its limitations](#) explain why it fails the exact prime-dilation laws of φ . The same section retains the cofinal LCM-diagonal criterion, its endpoint sign conditions and the limits of finite integer combinations of shifts.

The [signed interpolation argument](#) gives a different family of comparison sequences. For each integer $\beta \geq 2$, their sums can take any value in an interval about $\sum \varphi(n)\beta^{-n}$, while a prescribed prefix and every odd coefficient stay fixed. The errors may be bounded by any positive $f(n) \rightarrow \infty$, and the changed even indices have upper Banach density zero. On each fixed arithmetic progression, the difference of the associated Dirichlet series is entire. At roots of unity, the difference of the generating functions extends smoothly in the radial direction, with any prescribed finite collection of radial derivatives vanishing. Thus the Dirichlet singularities are preserved, although regular parts and zeros may change; finitely many radial derivatives do not determine the full radial germ. This ordinary proof has not been formalised in Lean and gives no cancellation estimate for the unchanged totient sequence.

Rational approximation imposes a different requirement. The identity

$$\Theta_2 := \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2} = S - \frac{1}{2}$$

was posted by Amiram Eldar, together with its coprimality interpretation, on 15 March 2026 [8], and by Steve Fan on 16 May 2026 [9]. Consider the rational quotients

$$Q(e, Y) = \frac{(\sum_{d \leq Y} \mu(d)/(2^d - 1)^{e+2})^2}{\sum_{d \leq Y} \mu(d)/(2^d - 1)^{2e+2}}.$$

Theorem B.3 (A lower bound for the rank-one quotients). *For $e \geq 1$ and $Y \geq 4$, the denominator of $Q(e, Y)$ is positive, and the unique minimiser is $(e, Y) = (1, 5)$. Every admissible quotient and every nonempty finite positive weighted average of such quotients satisfies*

$$Q - \Theta_2 > \frac{21}{320}.$$

The uniform bound $Q - \Theta_2 > 1/15$ is false already at $(1, 5)$.

The [proof](#) first makes the denominator uniformly positive. A geometric bound handles all $e \geq 2$; at $e = 1$, seven-term prefixes leave only $Y = 4, 5, 6$ to compare directly. The averaging assertion concerns averages of the quotients themselves. It leaves open signed combinations, quotients formed after combining the sums, and higher-rank kernels. A proposed family p_j/q_j in lowest terms would need $0 < q_j|\Theta_2 - p_j/q_j| \rightarrow 0$, which convergence alone does not imply.

C An exponential-sum question

For every $h \geq 1$ and every X_0 , suppose that some integers $X \geq \max(X_0, 1)$ and $L \geq h$ satisfy

$$16(2X + h + L + 2) \leq 2^L, \quad \operatorname{Re} \sum_{N=X}^{2X-1} \exp(2\pi i D_{h,N,L}/2^L) \leq \frac{9}{10}X. \quad (6)$$

These inequalities would prove irrationality. Indeed, if every residue failed (4), each phase would be within $1/16$ of an integer, so every real part would be at least $\cos(\pi/8) >$

9/10. Some N in the block must therefore satisfy (4). For a hypothetical rational value, we choose its period and then a sufficiently late block, contradicting (2).

To separate the counting estimates from the cancellation estimate, we single out one totient in each window. Fix $h, s \geq 1, L \geq s + h, X \geq 1$ and $0 < \eta < 1$, and put $t = L - s + 1$ and $E_N = \exp(2\pi i D_{h,N,L}/2^L)$. For $X \leq N < 2X$, let p_N be the largest prime factor of $N + t$ and $m_N = (N + t)/p_N$. Define

$$\begin{aligned} \mathcal{A} &= \{N \in [X, 2X) : m_N \leq \lfloor \sqrt{X}/2 \rfloor, p_N > 2\lfloor \sqrt{X} \rfloor\}, \\ \mathcal{G} &= \{N \in \mathcal{A} : \varphi(m_N) \geq \eta m_N\}. \end{aligned}$$

All indices are integers. On $\mathcal{A}$ we have $p_N > m_N$, so $\varphi(N + t) = \varphi(m_N)(p_N - 1)$. Its coefficient in the phase is $(2^h - 1)/2^t$. Write

$$z_N = \exp\left(\frac{2\pi i(2^h - 1)\varphi(m_N)(p_N - 1)}{2^t}\right), \quad w_N = E_N/z_N,$$

and let $\bar{z}_m$ be the average over the indices in $\mathcal{A}$ with cofactor m , taking the average to be zero for an empty set. We seek the four bounds

$$\begin{aligned} \operatorname{Re} \sum_{N \in \mathcal{G}} w_N(z_N - \bar{z}_{m_N}) &\leq \frac{14}{25}X, \\ \left| \sum_{N \in \mathcal{G}} w_N \bar{z}_{m_N} \right| &\leq \frac{1}{100}X, \\ \left| \sum_{N \in \mathcal{A} \setminus \mathcal{G}} E_N \right| &\leq \frac{1}{100}X, \\ \left| \sum_{\substack{X \leq N < 2X \\ N \notin \mathcal{A}}} E_N \right| &\leq \frac{8}{25}X. \end{aligned} \tag{7}$$

The last two sums cover the indices outside $\mathcal{G}$. On $\mathcal{G}$, the identity $w_N z_N = w_N(z_N - \bar{z}_{m_N}) + w_N \bar{z}_{m_N}$ gives the first two. The four sums therefore add to $\sum E_N$ before any estimates are taken, and their bounds give $(14/25 + 1/100 + 1/100 + 8/25)X = 9X/10$. Subtracting $\bar{z}_m$ cancels the unweighted sum over the whole cofactor group, but w_N varies with the same index N . That cancellation does not persist automatically after multiplication by w_N . Both of the first two sums still require weighted estimates; no independence has been assumed. The conclusion concerns the real part; it does not give a $21X/25$ bound for the absolute value of the full sum.

Problem C.1 (Four estimates for the first-harmonic sum). For each $h \geq 1$, do there exist $s \geq 1$ and $0 < \eta < 1$ such that, for every X_0 , there are X, L with

$$\max(X_0, 1) \leq X, \quad h \leq L - s, \quad 16(2X + h + L + 2) \leq 2^L,$$

for which those four bounds hold?

The parameters s and η may depend on h , but must be fixed before the threshold X_0 is given. All four estimates must then hold for one common choice of X, L . For each fixed h, s , take the least admissible L satisfying the size inequality in (6) and $L \geq s + h$. The [smooth-number and small-cofactor estimates](#) give the last two bounds for all sufficiently large X when $0 < \eta < 1/(300 \log 4)$. The elementary excluded-cofactor estimate is $(3\eta \log 4 + o(1))X$; it needs no prime number theorem. The first two bounds in (7) remain open. No optimality is asserted for the sufficient margins in (7).

Alternatively, at $s = 26$, $\eta = 1/1000$ and minimal depth one has $|G| > 67X/100$ for large X . The single open estimate $\text{Re} \sum_{N \in G} E_N \leq 603X/1000$, for each h on arbitrarily large blocks, would suffice by the same argument. The [ordinary comparison argument](#) gives larger first harmonics at all sufficiently large admissible scales, distinguishing its rational sequences from the cancellation required of φ . Its proof is included with the interpolation construction.

Balasubramanian, Giri and Srivastav give correlation estimates applicable to $\varphi(n)/n$, uniformly for $|h| \leq x/2$ [11, Thm. 2.2]. Here the modulus grows with L , and the residual weight depends on other totient values in the window. That correlation theorem supplies neither of the two remaining estimates. Establishing them for the actual totient is the unresolved arithmetic step.

Verification and reproducibility

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The [verification record](#) gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded. The [declaration guide](#) links the formal sources for this paper. Complete proofs of three cited results appear in Appendix A of the companion paper. The displayed arguments, including signed interpolation, require ordinary mathematical review; the recorded Lean and Comparator checks cover their specified declarations and source versions. No new formal run or independent review was performed for this revision.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.1	Lean · Comparator	Corollary 3.1	Lean · Comparator
Theorem 4.1	Lean · Comparator	Lemma 4.2	Lean · Comparator
Corollary A.1	Lean · Comparator	Corollary A.2	Lean · Comparator
Theorem B.1	Lean · Comparator	Theorem B.3	Lean · Comparator